A PROJECT REPORT ON

HACKAGON

A Web Platform for Hosting and Joining Online Coding Competitions

INDIVIDUAL CONTRIBUTION REPORT

Member 1: Accounts and Rooms

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Computer Science and Engineering

Submitted by

Hiten Sharma

(Member 1 of the HackAgon team: Hiten Sharma, Harsh Sharma, Hemanth Singh Tanwar, Arjun Suman, Akshat Rathore)

Under the guidance of

Er. Ram Babu Buri

Department of Computer Science and Engineering
Arya College of Engineering and IT, Jaipur

Session 2026-27

CERTIFICATE

This is to certify that the individual contribution report entitled “HackAgon: Individual Report of Member 1 (Accounts and Rooms)”, submitted by Hiten Sharma, a student of the third year of Bachelor of Technology in Computer Science and Engineering at Arya College of Engineering and IT, Jaipur, is a record of his own work carried out during the session 2026-27 as a member of the HackAgon project team, under my guidance and supervision.

The work embodied in this report has not been submitted elsewhere for the award of any degree or diploma. The report is, in my opinion, worthy of consideration as part of the project submission.

Er. Ram Babu Buri	**Dr. Akhil Pandey**
Project Guide	Head of Department

Date: _____

Place: Jaipur

DECLARATION

I, Hiten Sharma, hereby declare that the individual contribution report entitled “HackAgon: Individual Report of Member 1 (Accounts and Rooms)” is an original account of the work done by me as Member 1 of the HackAgon project team, under the guidance of the project guide named in the certificate.

The report is based on my own work, supported by the HackAgon Daily Work Log of the team (6 July 2026 to 6 October 2026) and the HackAgon Team Modules and 12-Week Plan. It has not been copied from any other source. Wherever the work of other team members is described, it is attributed to them by name or by member number.

Hiten Sharma

Date: _____

Place: Jaipur

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who helped me complete my part of the HackAgon project. I thank our project guide for constant direction, patient review and helpful suggestions throughout the project. I am grateful to the Head of the Department of Computer Science and Engineering and to the faculty of Arya College of Engineering and IT, Jaipur, for the learning environment and the laboratory facilities.

I thank my teammates Harsh Sharma, Hemanth Singh Tanwar, Arjun Suman and Akshat Rathore. The accounts and rooms could only be useful because their modules, the judge and admin tools, the money and database core, the proctoring and host dashboard, and the design and learning screens, were built in the same weeks and fitted together with mine. Their reviews of my pull requests and my reviews of theirs taught me as much about collaboration as about code.

I also thank the open-source community. HackAgon stands on Java, Spring Boot, MySQL, Render, GitHub and many small libraries, all of which are free to use and well documented. Finally, I thank my friends and family for their encouragement.

Hiten Sharma

ABSTRACT

HackAgon is a web application for hosting and joining online coding competitions, with solo, duo, squad and duel rooms, paid tournaments with a funded prize pool, automatic judging, proctoring and an administrator back office. It is built by a team of five students. This report describes the contribution of Member 1, Hiten Sharma, who owned two modules: Module 1, Accounts and OTP, and Module 2, Rooms and matches.

For Module 1, the report covers the audit of the existing authentication code, email OTP with a 5 minute expiry, 5 attempts and a 60 second resend, SMS OTP through MSG91, forgot password, profile edit with re-verification and the age rule for paid tournaments. For Module 2, it covers the room data model, the room create API with editable defaults, the server timer for room status, joining and starting, the submission endpoint with the judge call and on-time check, ranks and tie-break, profile counters from real records, tournament mode with consent and lock refusal, elimination, result locking and organisation hosting. It then describes the hardening and launch work of weeks 10 to 14: the end-to-end test script, keyboard and screen-reader checks, the legal pages with the signup checkbox, launch-week monitoring and the fixes after the first real tournament.

A full chapter is devoted to collaboration. I worked with Member 2 on the submission path and disputes, with Member 3 on the data model, payouts, security and deployment, with Member 4 on locks and elimination, and with Member 5 on the pages and the match screen. In total I logged 194.0 hours over 67 working days, from 6 July to 6 October 2026. Every feature in the report is traced to a dated entry in the daily work log.

Keywords: accounts, one-time password, rooms, matches, tournament mode, result locking, accessibility, launch, collaboration, Java, Spring Boot, MySQL.

CONTENTS

1. INTRODUCTION..... 1
1.1 About the HackAgon Project..... 1
1.2 Purpose of This Report..... 1
1.3 Member 1 at a Glance..... 1
1.4 Sources and Method..... 2
1.5 Organisation of the Report..... 2
2. ROLE AND RESPONSIBILITIES..... 3
2.1 The Two Modules..... 3
2.2 Files, Pages and Responsibilities..... 3
2.3 Decisions Owned by Member 1..... 4
2.4 What Belonged to Other Members..... 4
2.5 Effort Summary..... 4
3. REQUIREMENTS AND DESIGN..... 7
3.1 Requirements Covered..... 7
3.2 Non-Functional Requirements That Shaped the Work..... 8
3.3 Position in the Architecture..... 8
3.4 Data Design..... 9
3.5 Design Principles..... 9
4. MODULE 1: ACCOUNTS AND OTP..... 10
4.1 Audit of Accounts (Week 1)..... 10
4.2 Email OTP (13 and 14 July)..... 10
4.3 SMS OTP with MSG91 (15 July)..... 11
4.4 Forgot Password (16 July)..... 12
4.5 Profile Edit with Re-verification (17 July)..... 12
4.6 The Age Rule (19 August)..... 12
4.7 Pages and Copy..... 12
5. MODULE 2: ROOMS AND MATCHES..... 13
5.1 Room and Match Data Model (9 and 10 July)..... 13
5.2 Room Create API (20 July)..... 13
5.3 Editable Defaults and the Create-room Page (21 and 22 July)..... 13
5.4 Room Status Timer (23 July)..... 13
5.5 Join and Start (27 and 28 July)..... 14
5.6 Submission Endpoint (29 and 30 July)..... 14
5.7 Testing a Full Solo Room (31 July)..... 14
5.8 Results, Ranks and Profile Counters (3 to 6 August)..... 14
5.9 Tournament Mode and Consent (10 to 14 August)..... 15
5.10 Elimination and the Paid-tournament Age Rule (17 to 20 August)..... 15
5.11 Result Locking (24 to 28 August)..... 15
5.12 Organisation Hosting (31 August to 4 September)..... 16
6. HARDENING, LEGAL PAGES AND LAUNCH..... 17
6.1 End-to-end Test Script (7 to 9 September)..... 17
6.2 Accessibility Checks (10 and 11 September)..... 17

6.3 Legal Pages and the Acceptance Checkbox (14 to 18 September)....17
6.4 Code Freeze and Launch Week (21 to 25 September)....17
6.5 After Launch (28 September to 2 October)....18
6.6 Final Report and Submission (5 and 6 October)....18
7. COLLABORATION WITH OTHER MEMBERS....19
7.1 Overview of Dependencies and Joint Functions....19
7.2 Collaboration with Member 2 (Harsh Sharma)....20
7.3 Collaboration with Member 3 (Hemanth Singh Tanwar)....20
7.4 Collaboration with Member 4 (Arjun Suman)....21
7.5 Collaboration with Member 5 (Akshat Rathore)....21
7.6 Team-Wide Activities....21
7.7 Summary of Collaboration....21
8. TESTING AND VERIFICATION....23
8.1 Test Strategy....23
8.2 Tests Recorded in the Work Log....23
8.3 Bug-fix Days....23
8.4 Verification by Other Members....24
8.5 Points to Note....24
9. WEEK-BY-WEEK SUMMARY....25
9.1 Weeks 1 and 2: Audit and Accounts....26
9.2 Weeks 3 to 5: Rooms, the First Playable Match and Results....26
9.3 Weeks 6 to 9: Tournament Mode, Host Controls, Result Locking and Organisations....27
9.4 Weeks 10 to 12: Hardening, Legal Pages and Launch....27
9.5 Weeks 13 and 14: After Launch and Handover....27
10. CHALLENGES, DECISIONS AND LIMITATIONS....28
10.1 Challenges and How They Were Met....28
10.2 Delivered Features....28
10.3 Notes on the Sources....29
10.4 Limitations....29
11. CONCLUSION AND FUTURE SCOPE....30
11.1 Conclusion....30
11.2 Lessons Learnt....30
11.3 Future Scope....30
REFERENCES....31
APPENDIX A: COMPLETE DAILY WORK LOG OF MEMBER 1....32
APPENDIX B: DATED JOINT WORK WITH OTHER MEMBERS....35

LIST OF FIGURES

Figure 2.1: Hours logged by Member 1 per kind of work.....5
Figure 2.2: Timeline of Member 1’s work across the fourteen logged weeks.....6
Figure 2.3: Share of logged hours by project stage.....6
Figure 3.1: Layered architecture around Member 1’s modules.....8
Figure 4.1: OTP request, delivery and verification with its limits.....11
Figure 4.2: Sequence diagram of OTP request and verification.....11
Figure 5.1: Room status, join, submission and result steps owned by Member 1.....13
Figure 5.2: Sequence of a code submission through Member 1’s endpoint and Member 2’s judge 14
Figure 5.3: Result locking: confirmation, auto-confirmation, dispute and hand-over to payouts…16
Figure 7.1: Collaboration map of Member 1 with the other four members.....19
Figure 7.2: Joint work items per partner (from Appendix B).....22
Figure 9.1: Cumulative hours logged by Member 1 week by week.....26
Figure 9.2: Hours logged per day, 6 July to 6 October 2026 (one bar per weekday).....26

LIST OF TABLES

Table 1.1: Summary of Member 1 1
Table 2.1: Files and pages of Member 1 3
Table 2.2: Hours by area of work 4
Table 3.1: Requirements and Member 1’s contribution 7
Table 3.2: Data used by Member 1’s modules 9
Table 4.1: Week 1 activities of Member 1 10
Table 6.1: Work of weeks 10 to 14 18
Table 7.1: Matrix of joint functions 19
Table 8.1: Tests carried out by Member 1 23
Table 9.1: Weekly summary of Member 1’s work 25
Table 10.1: Challenges and responses 28
Table 10.2: Status of Member 1’s features 28
Table A.1: Daily work log of Member 1 32
Table B.1: Dated joint work 35

1. INTRODUCTION

1.1 About the HackAgon Project

HackAgon is a web platform for hosting and joining online coding competitions. A college club, a class or a group of friends can create a room in a few minutes, pick questions, invite players and, when they want stakes, run a paid tournament in which the host funds the prize pool in advance and the pool is paid out by a published split after a flat 1% platform fee. The application is built on Java and Spring Boot with a MySQL database, a set of plain HTML pages, and shared browser scripts and style files.

The project was carried out by a team of five students of Arya College of Engineering and IT, Jaipur. The work was divided into ten modules, and every member owns modules of equal total weight. This report is about **Member 1, Hiten Sharma**, whose area in the team plan is **Accounts and Rooms**.

1.2 Purpose of This Report

The team plan lists the modules and the weekly tasks of all five members. The daily work log records what each member actually did, day by day. Neither document describes one member’s contribution in full. This report collects in one place everything I did: what I built, in which week, how it was tested and how it connected to the work of the other four members.

HackAgon is a tightly connected system. An account created in my module is used by the payout details of Member 3, a room created in my module is judged by the code of Member 2, a lock set by Member 4 is enforced on my match page, and the final rank list produced by my module feeds the payouts of Member 3. Wherever I worked together with another member on a shared function, the report names the shared function, the interface between the two sides and the dates on which each side did its part.

1.3 Member 1 at a Glance

Table 1.1: Summary of Member 1

Item	Details
Name	Hiten Sharma (Member 1)
Role in the team plan	Accounts and Rooms
Modules owned	Module 1: Accounts and OTP; Module 2: Rooms and matches
Main files	`AuthController.java, SecurityConfig.java, JwtService.java, OtpService.java (with mail delivery), RoomsController.java, PlayController.java (shared), DomainService.java`

Item	Details
Main pages	login, signup, verify-otp, profile, create-room, arena, match (the match page is shared with Member 5)
Period covered by the log	6 July 2026 to 6 October 2026 (weekdays), project weeks 1 to 14
Work logged	67 weekdays, all marked Completed; **194.0 hours** (2 to 4 hours a day; 14.5 hours in each of weeks 1 to 13 and 5.5 hours in the two-day week 14)
Collaborated with	Member 2 (Harsh Sharma), Member 3 (Hemanth Singh Tanwar), Member 4 (Arjun Suman), Member 5 (Akshat Rathore)

1.4 Sources and Method

This report is based on my own work, supported by two team documents: the HackAgon Team Modules and 12-Week Plan, and the HackAgon Daily Work Log. The plan supplies the module descriptions, the owners, the file names, the weekly tasks and the risks. The log supplies the day-by-day record of what each member did, including my 67 entries and the entries of the other four members that touch my work. Dates follow the day-month-year format of the log.

The plan is the authority on what each task was meant to deliver and who owned it. The log is the authority on what I actually did and on which date, so every date in this report comes from the log. The plan sets out twelve weeks of tasks; the log runs for fourteen project weeks from 6 July to 6 October 2026. Weeks 10 to 12 of the plan (the end-to-end test, accessibility checks, legal pages and launch monitoring) all appear in the log as completed, and weeks 13 and 14 cover the review after the first real tournament and the preparation of this report.

1.5 Organisation of the Report

Chapter 2 explains my role, the files and the effort. Chapter 3 lists the requirements and the design that the two modules follow. Chapter 4 describes Module 1, accounts and OTP, and Chapter 5 describes Module 2, rooms and matches. Chapter 6 covers the work of weeks 10 to 14: testing end to end, accessibility, legal pages, launch and the weeks after launch. Chapter 7 is devoted to collaboration with the other members. Chapter 8 reports testing, Chapter 9 gives a week-by-week account, and Chapter 10 discusses challenges, decisions and limitations. Chapter 11 concludes with lessons learnt and future scope. Appendix A reproduces my complete daily log and Appendix B lists the dated joint work with the other members.

2. ROLE AND RESPONSIBILITIES

2.1 The Two Modules

In the team plan each member owns two modules, one larger and one smaller, so that the load is shared evenly. I own the two modules that decide who can enter the platform and how a contest is created, joined and finished.

Module 1: Accounts and OTP (weight 2)

This module covers signup, login, JWT sessions, email and SMS one-time passwords, forgot password, profile details and the age check. Its files are `AuthController.java`, `SecurityConfig.java`, `JwtService.java`, `OtpService.java (with mail delivery)`, together with the login, signup, verify-otp and profile pages.

Module 2: Rooms and Matches (weight 3)

This module covers the room creation form, the question side panel, the Arena list, joining, the live match page, scoring, the scoreboard and the results. Its files are `RoomsController.java`, `PlayController.java` and `DomainService.java`, together with the create-room, arena and match pages. The weights of 2 and 3 give me a load of 5 points, the same as every other member.

2.2 Files, Pages and Responsibilities

Table 2.1: Files and pages of Member 1

File or page	Responsibility
`AuthController.java`	Signup, login, OTP request and verify, forgot password, profile update
`SecurityConfig.java`, `JwtService.java`	Password hashing, JWT creation and checking, cookie settings
`OtpService.java`	Creating, storing and checking one-time passwords, expiry, attempt count and resend timing
`OtpService.java (mail delivery)`	Sending email OTP. Mail first went through Gmail; Member 3 moved it to a transactional provider on 11 September
`RoomsController.java`	Room create API with validation and defaults, room status, join, start, host actions such as eliminate and confirm results
`PlayController.java` (shared)	Submission endpoint written by me, which calls the judge of Member 2; the hook for saving wrong submissions was agreed with Member 5
`DomainService.java`	Shared rules of the domain: ranks, tie-break, on-time check, counters
login, signup, verify-otp, profile	Account pages; styled by Member 5, wired to the API by me
create-room, arena, match	Room pages; designed by Member 5, connected to the room and play APIs by me

2.3 Decisions Owned by Member 1

- **The server decides, not the browser.** OTP expiry, attempt limits, resend timing, room status changes and the on-time check of a submission are all decided on the server.
- **Real counters only.** Profile counters such as rooms played and rooms won are calculated from real records, so a new account shows zero everywhere, as the product requirements demand.
- **Verification on change.** Changing the email or the mobile number on the profile asks for a new OTP before the change is saved.
- **Age rule.** Only users aged 18 or over may enter or host paid tournaments.
- **No silent results.** A result is confirmed by the host, or confirmed automatically after 24 hours if there is no dispute.

2.4 What Belonged to Other Members

Clear boundaries help the team avoid overlap. I did not own the code judge or the question bank (Member 2), the money tables, payouts, deployment and the database core (Member 3), the proctoring endpoints and the host dashboard (Member 4), or the design system and the learning screens (Member 5). My code calls or feeds these parts at agreed points, which are described in Chapter 7.

2.5 Effort Summary

I logged work on all 67 weekdays between 6 July and 6 October 2026, a total of **194.0 hours**. Daily hours ranged from 2 to 4, every full week came to 14.5 hours, and the last week (5 and 6 October) came to 5.5 hours. All entries are marked Completed. Table 2.2 groups the hours by area of work; each of the first twelve areas is exactly one project week.

Table 2.2: Hours by area of work

Area of work	Weeks	Days	Hours	Share
Account audit and data model	1	5	14.5	7.5%
Accounts and OTP	2	5	14.5	7.5%
Room creation	3	5	14.5	7.5%
Join, submit and play	4	5	14.5	7.5%
Results and profile counters	5	5	14.5	7.5%
Tournament mode and consent	6	5	14.5	7.5%
Elimination and age rule	7	5	14.5	7.5%
Result locking	8	5	14.5	7.5%
Organization hosting	9	5	14.5	7.5%

Area of work	Weeks	Days	Hours	Share
End-to-end test and accessibility	10	5	14.5	7.5%
Legal pages	11	5	14.5	7.5%
Launch and monitoring	12	5	14.5	7.5%
Post-launch fixes and handover	13 to 14	7	20.0	10.3%
Total	**1 to 14**	**67**	**194.0**	**100%**

The work in the log falls into six kinds of activity. Building new features took more than half of the logged time. Testing (including the end-to-end script and the accessibility checks) took about a sixth, and documentation, bug fixing, monitoring and review share the rest. Figure 2.1 shows the split.

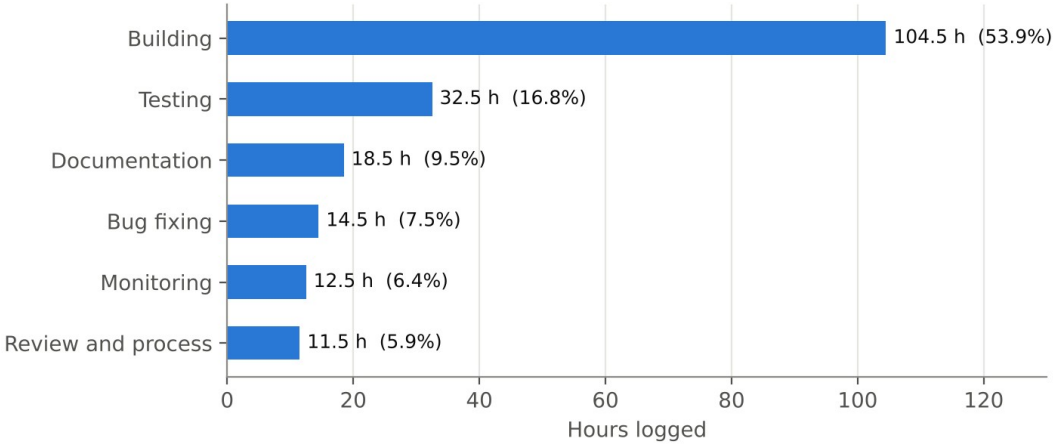


Figure 2.1: Hours logged by Member 1 per kind of work

Figure 2.2 shows when each area of work took place. The account audit and the OTP flows fill the first two weeks, rooms and the first playable match follow in weeks 3 to 5, and tournament mode, host controls, result locking and organisation hosting fill weeks 6 to 9. Weeks 10 to 12 are the hardening and launch weeks, and weeks 13 and 14 close the project.

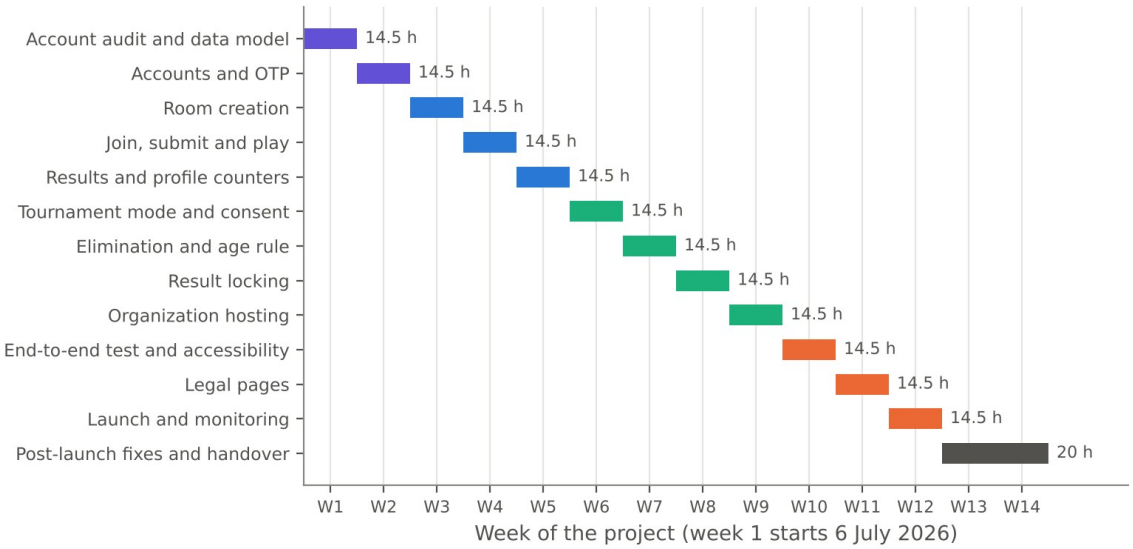


Figure 2.2: Timeline of Member 1's work across the fourteen logged weeks

Figure 2.3 groups the same hours into five stages of the project. The largest stage is weeks 6 to 9, because tournament mode and elimination touch the match page, the consent page, the age rule and the host controls, and result locking connects the match to the payouts.

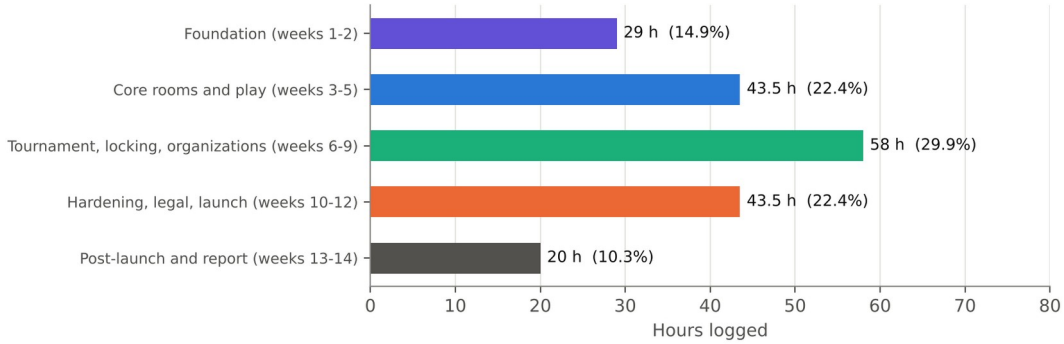


Figure 2.3: Share of logged hours by project stage

3. REQUIREMENTS AND DESIGN

3.1 Requirements Covered

The plan states what each module must do. Table 3.1 lists the requirements that my tasks address and the contribution made.

Table 3.1: Requirements and Member 1’s contribution

Requirement (from the plan)	Contribution and date
Email OTP with 5 minute expiry, 5 attempts and 60 second resend	Built in week 2 (13 and 14 July)
SMS OTP	Added using MSG91 (15 July)
Forgot password	Built with OTP (16 July)
Profile edit asks for a new OTP when email or mobile changes	Built (17 July)
Room create API with validation and the PRD’s editable defaults	Built and wired to `create-room.html` (20 to 22 July)
Room status open, live, finished run by a timer on the server	Added (23 July)
Join by Arena or invite code; start the match	Built (27 and 28 July)
Server decides whether a submission is on time	Added with the judge call (29 and 30 July)
Final ranks with tie-break; results page data	Built (3 and 4 August)
Profile counters from real records; new account shows zero	Calculated and tested (5 and 6 August)
Tournament mode, consent page, locked players refused on every request	Built and tested (10 to 13 August)
Host eliminates a player with a reason; no rejoin; removed from ranking	Built (17 and 18 August)
Only users aged 18 or over can enter or host paid tournaments	Added (19 August)
Host confirms results or they auto-confirm after 24 hours; rank list to money module	Built (24 to 26 August)
Organisation host application, admin approval, higher limits	Built (31 August to 3 September)
End-to-end script of signup, room creation, join, solve and results	Written, extended and run (7 to 9 September); re-run on 30 September
Keyboard and screen-reader check of auth and room pages	Done (10 and 11 September)
Terms of service, privacy policy, refund policy, webcam consent text, acceptance checkbox at signup	Written and added (14 to 18 September)
Launch-week monitoring of signups, OTP delivery, login errors and the first real tournament	Done (22 to 24 September)

3.2 Non-Functional Requirements That Shaped the Work

- **Security of accounts.** Passwords are hashed with BCrypt, sessions use JWT with an expiry, and cookie flags are checked. Week 1 was spent auditing these settings.
- **Server authority.** Anything that affects fairness or money, such as time, locks, eliminations and results, is checked on the server for every request.
- **Honest numbers.** No demo users, scores or counters are placed in the app; counters start at zero and grow from real records.
- **Clear messages.** Every feature has hand-written empty and error messages, which is part of the team's definition of done.
- **Deployability.** The main branch is always deployable. It went to a staging service on Render during the build and to a separate production service from 22 September.

3.3 Position in the Architecture

The application is a Java and Spring Boot server with a MySQL database through the MySQL Connector/J (JDBC) driver. The browser pages talk to the server through js/api.js. My routes sit in the controller package (auth, rooms and play), and the shared logic sits in the service package (security, JWT, OTP and domain services). The libraries used by my modules are Spring Boot (Web, Security and JDBC starters), MySQL Connector/J and JJWT, with MSG91 for SMS. Figure 3.1 shows the layers and where my code sits.

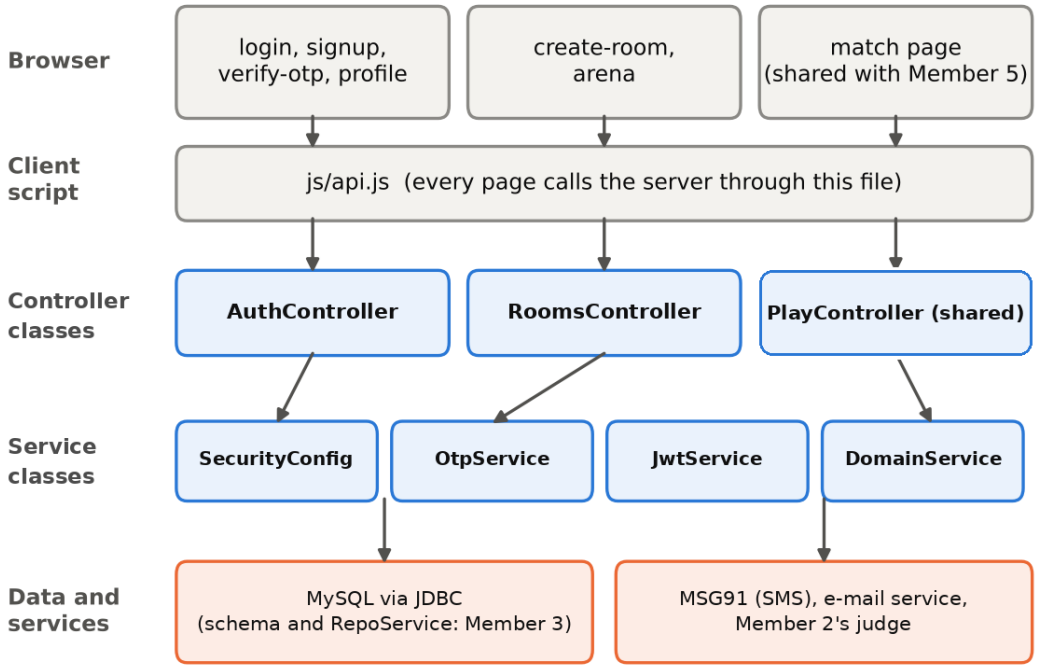


Figure 3.1: Layered architecture around Member 1's modules

3.4 Data Design

In week 1 I drafted the room and match data model and reviewed it with Member 3, who owns `schema.sql` and the repository layer. The model describes the following ideas. The exact table and column names are those of `schema.sql` and are not repeated here.

Table 3.2: Data used by Member 1's modules

Data	Purpose
Users	Name, email, mobile, password hash, date of birth for the age check, verification state
OTP records	Code, purpose, expiry time, attempts used, time of last send
Rooms	Settings from the creation form, mode (solo, duo, squad, duel, tournament), status, host, start and end times
Room players	Who joined which room, elimination flag and reason, lock state (written by Member 4's endpoints)
Submissions	Language, question, verdict, time and memory (written with Member 2's judge)
Results	Final ranks, confirmation state, auto-confirm deadline

3.5 Design Principles

- **Small steps with a test each.** Almost every building week in the log ends with a testing day and a bug-fix day.
- **Contracts before connections.** Where two modules meet, such as the money module, the owners agree the request and response first.
- **Defaults that the host can edit.** The room form starts from the PRD defaults and lets the host change them, which keeps room creation fast.
- **Fail closed.** A locked or eliminated player is refused on every request; an unverified contact change is not saved.

4. MODULE 1: ACCOUNTS AND OTP

4.1 Audit of Accounts (Week 1)

Week 1 was an audit. Instead of starting from nothing, I read the existing authentication code and compared it with the product requirements. The team plan notes that existing pages and routes count as a first skeleton, so the early weeks make it real.

Table 4.1: Week 1 activities of Member 1

Date	Hours	Activity
06-07-2026	3	Read AuthController.java and SecurityConfig.java, JwtService.java to understand login flow
07-07-2026	2.5	Checked password hashing and JWT expiry settings
08-07-2026	4	Checked cookie flags and listed gaps against the PRD
09-07-2026	3	Drafted room and match data model
10-07-2026	2	Reviewed data model with Member 3 and noted changes

The result of the audit was a list of gaps against the PRD. That list became the task list of weeks 2 and 3: the missing OTP rules, SMS delivery, forgot password and re-verification on profile changes.

4.2 Email OTP (13 and 14 July)

I built the email one-time password with a 5 minute expiry on 13 July. On 14 July I added the attempt limit of 5 and the resend delay of 60 seconds. An attempt counter stops guessing, the expiry stops old codes from being used, and the resend delay stops a user from flooding an inbox. Mail is sent through `OtpService.java`. During the build the mail went out through Gmail; Member 3 later moved it to a transactional provider on 11 September, which needed no change to the OTP rules.

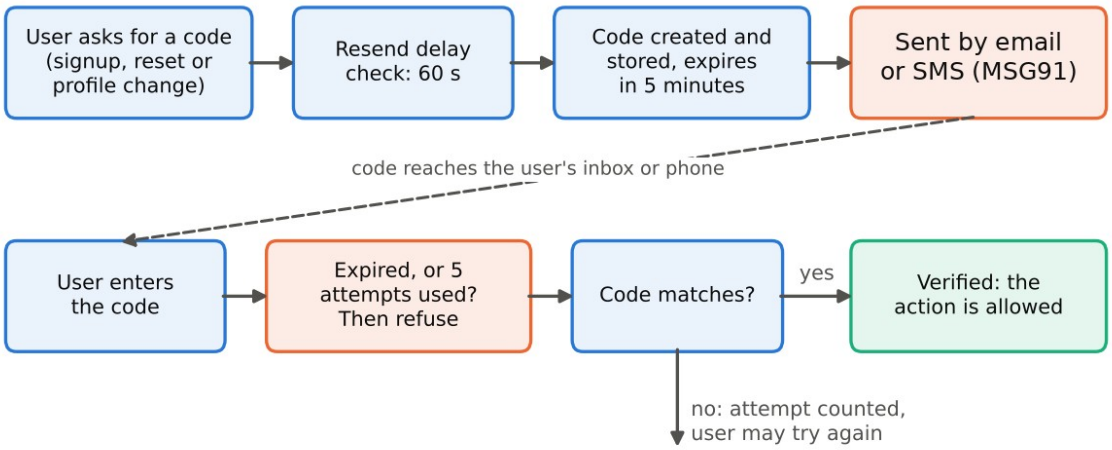


Figure 4.1: OTP request, delivery and verification with its limits

Figure 4.2 shows the same flow as a sequence between the user, the browser, the auth route, the OTP library and the delivery service.

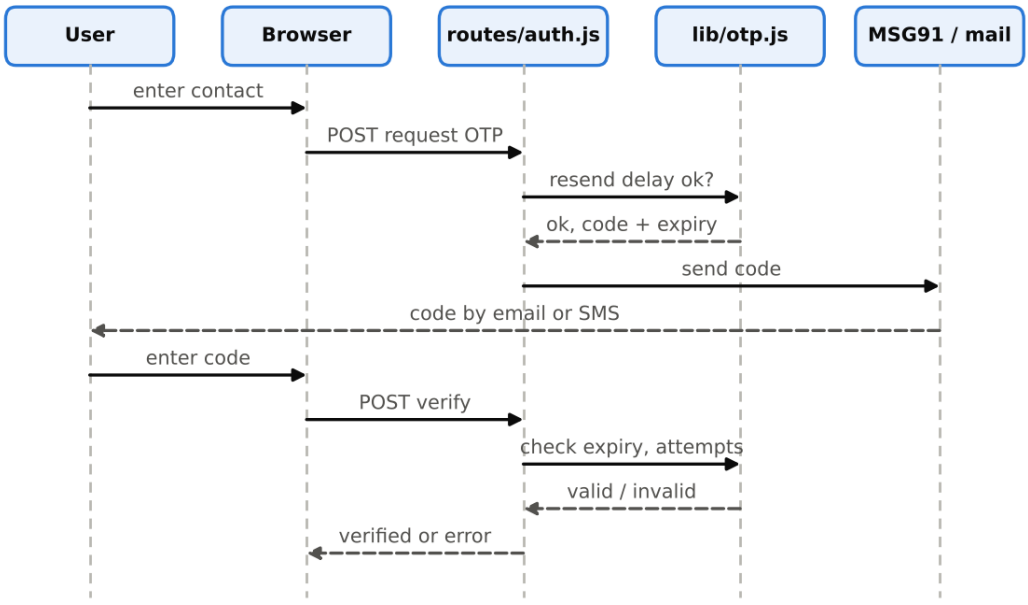


Figure 4.2: Sequence diagram of OTP request and verification

4.3 SMS OTP with MSG91 (15 July)

On 15 July I added an SMS OTP using MSG91. The same rules of expiry, attempts and resend apply, so email and SMS share one OTP library. Sharing the library means a rule is changed in one place and applies to both channels.

4.4 Forgot Password (16 July)

The forgot password flow uses the same OTP. A user asks for a reset, receives a code, proves control of the contact by entering it, and then sets a new password. Using one OTP library for signup, reset and profile change keeps the rules identical in every place.

4.5 Profile Edit with Re-verification (17 July)

A profile edit that changes the email address or the mobile number asks for a new OTP to the new contact before it is saved. This protects the account, and it also protects Member 3's payout details, which send an OTP to the profile mobile number (24 August). If a mobile number could be changed without proof, that payout check would be weak.

4.6 The Age Rule (19 August)

On 19 August I added the rule that only users aged 18 or over may enter or host paid tournaments. The rule lives with the account because it depends on the date of birth in the profile, and it is checked on the server when a paid tournament is joined or created. It sits between Member 3's entry fee checkout (17 August) and the join after payment (18 August) on one side and the refund work (19 and 20 August) on the other.

4.7 Pages and Copy

Member 5 styled the login, signup and verify-otp pages with hand-written copy on 17 July, the same day that I finished profile re-verification. I supplied the API behaviour and the error cases, and Member 5 wrote the messages and the layout. The acceptance checkbox for the terms and privacy policy was added to the signup page on 18 September (Section 6.3).

5. MODULE 2: ROOMS AND MATCHES

5.1 Room and Match Data Model (9 and 10 July)

I drafted the model on 9 July and reviewed it with Member 3 on 10 July. Member 3 fixed the version 1 tables on 13 July, finished the migration runner on 14 July and the repository layer on 15 July, so the model went into the schema without delay.

5.2 Room Create API (20 July)

The room create API validates every field before a room is stored. Validation keeps wrong values out of the database and returns clear messages to the form. Member 3's input validation and error handler (21 and 23 July) work around the same routes.

5.3 Editable Defaults and the Create-room Page (21 and 22 July)

On 21 July I added the PRD default values to the room form, with each default editable by the host. On 22 July I connected `create-room.html` to the API. Member 5 had already designed the page and its question side panel on 20 July and the topic and difficulty filters on 21 July, so the page was ready to connect. The filter API that feeds the side panel was built by Member 2 on 24 July. Author to add: the PRD default values used in the room form.

5.4 Room Status Timer (23 July)

A server-side timer moves a room from open to live to finished. The browser does not decide when a room starts or ends. This makes every player see the same status and prevents a player from extending the contest by changing a local clock.

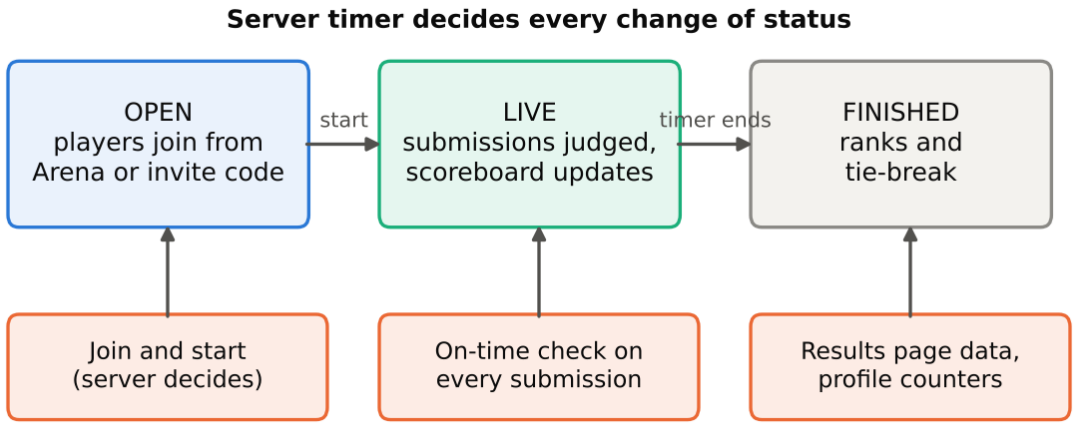


Figure 5.1: Room status, join, submission and result steps owned by Member 1

5.5 Join and Start (27 and 28 July)

Players join a room from the Arena list or with an invite code. On 28 July I added the start match logic. Joining and starting are decided on the server, not in the page. Member 5 had designed the Arena room cards on 22 July and built the match page in the same week: editor and question list on 27 July, timer on 28 July and the polling scoreboard on 29 July.

5.6 Submission Endpoint (29 and 30 July)

The submission endpoint in `PlayController.java` receives the code, the language and the question, calls the judge, and records the result. The judge itself belongs to Member 2, who built the submit flow, the test-case run and the verdict storage on 13 to 15 July. My endpoint adds the server-side on-time check, so a submission that arrives after the deadline is not scored. Member 2 limited concurrent judge runs (27 July) and allowed one pending submission per player per question (28 July), which my endpoint relies on.

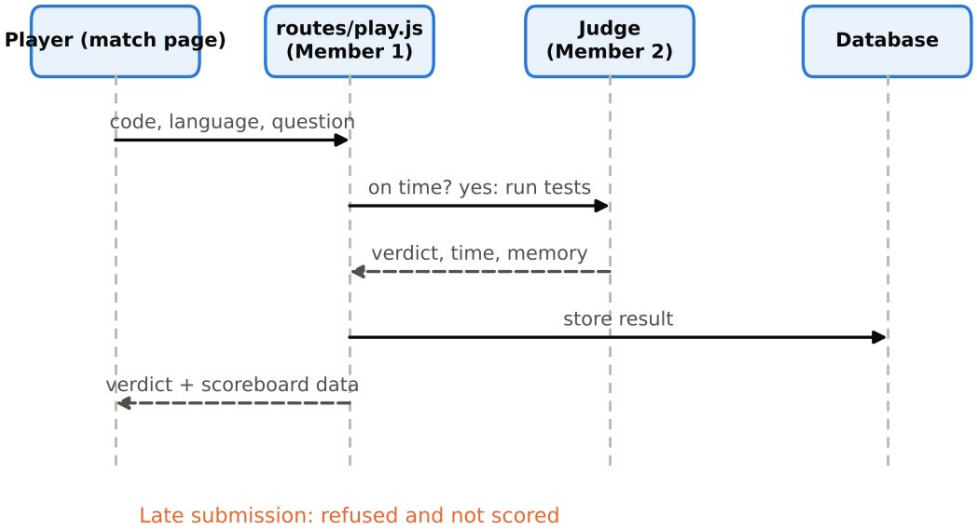


Figure 5.2: Sequence of a code submission through Member 1's endpoint and Member 2's judge

5.7 Testing a Full Solo Room (31 July)

On 31 July I tested a full solo room with two users. This was the test for Milestone 1 in the plan: two real users play a full solo room on staging. Member 3 deployed Milestone 1 to staging on 30 July and ran the test checklist on 31 July, and Member 5 fixed visual bugs on the match page the same day.

5.8 Results, Ranks and Profile Counters (3 to 6 August)

On 3 August I added the final rank rule and the tie-break. On 4 August I built the results page data, and on 5 August I calculated the profile counters, rooms played and rooms won, from real records. The test of 6 August checked that a new account shows zero everywhere,

which is the product requirement for honest numbers. Member 5's achievements (17 to 20 August) calculate their metrics from the same real counts. Author to add: the tie-break order used in DomainService.java.

5.9 Tournament Mode and Consent (10 to 14 August)

On 10 August I added the tournament mode option to room creation. On 11 August I built the consent page before joining, because tournament mode uses the webcam and the tab lock. Member 5 designed the consent screen on 10 August, the camera check screen on 11 August and the lock screen on 12 August, so the screens were ready as the logic was built. On 12 August I made the match page refuse locked players, and on 13 August I tested the lock check on every request. The locks themselves are set by Member 4's code: the lock-state endpoint (15 July), the camera permission and face check (10 August), snapshots (11 August), the server-side tab and minimize lock (12 August) and the 30 second network grace period (13 August).

5.10 Elimination and the Paid-tournament Age Rule (17 to 20 August)

A host can eliminate a player with a stated reason (17 August). An eliminated player cannot rejoin, and the score of that player is removed from the ranking (18 August). Member 4 built the live player list on 17 August and the eliminate button on 20 August, which calls this server logic. The 18+ rule was added on 19 August (Section 4.6), and both were tested on 20 August.

5.11 Result Locking (24 to 28 August)

The host confirms the results with a button (24 August), or the results confirm automatically after 24 hours if there is no dispute (25 August). A dispute raised by a player blocks the payout; Member 2 built the dispute form (17 August), the payout block (18 August), the admin review screen (19 August) and the decide and close actions (20 August). After confirmation, I sent the final rank list to the money module (26 August). On 27 August I tested result locking with Member 3, whose payout job deducts the 1% fee and splits the prize by rank.

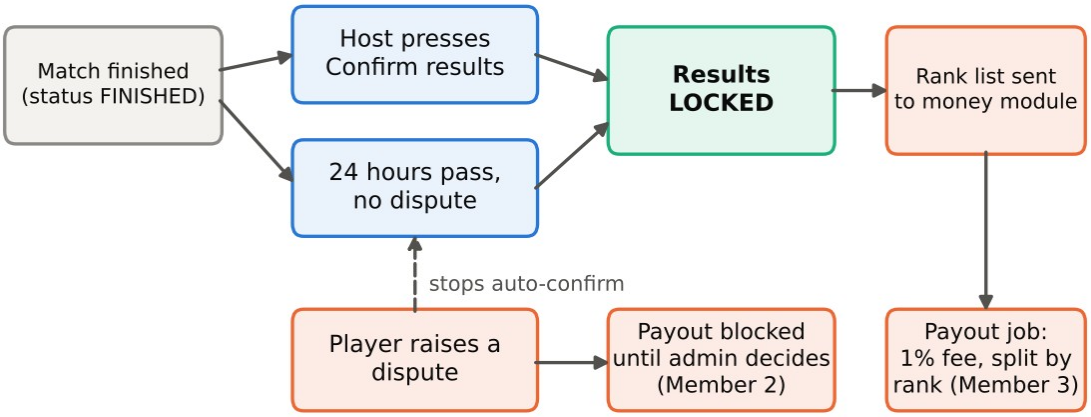


Figure 5.3: Result locking: confirmation, auto-confirmation, dispute and hand-over to payouts

5.12 Organisation Hosting (31 August to 4 September)

I built the organisation host application form (31 August), the admin approval flow (1 September), the higher limits for approved organisations (2 September) and the final profile page (3 September). The whole flow was tested end to end on 4 September. The approval screen sits in the admin area of Member 2, whose final admin dashboard layout was built on 31 August and whose panels were added on 1 and 2 September.

6. HARDENING, LEGAL PAGES AND LAUNCH

Weeks 10 to 14 turned the finished features into a launched product. This chapter covers the work of those five weeks, which the plan describes as testing, legal pages, launch and the weeks after launch.

6.1 End-to-end Test Script (7 to 9 September)

On 7 September I wrote a script that walks through signup, room creation and join. On 8 September I extended it to solve a question and show the results, so that it covers the whole path from a new account to a result. On 9 September I ran the script and fixed what broke. The script stands in for a user and can be run again after any change; I re-ran it on 30 September after the first tournament. In the same week Member 2 ran the judge load test (7 to 9 September) and Member 3 simulated 200 and then 500 players polling one room (7 and 8 September), and a few days later Member 2 tested the whole app as a stranger would, from signup to match (14 September).

6.2 Accessibility Checks (10 and 11 September)

On 10 September I did a keyboard check of the auth pages, and on 11 September a screen-reader check of the room pages. Member 5 worked on the same goal from the design side: colour contrast of at least 4.5:1 (7 and 8 September), clear focus states on buttons and inputs (9 September) and a final accessibility re-check on 5 October. The log records that the checks were done; it does not list the individual findings.

6.3 Legal Pages and the Acceptance Checkbox (14 to 18 September)

I wrote the terms of service on 14 September, the privacy policy on 15 September, the refund policy on 16 September and the webcam consent text on 17 September. On 18 September I added the acceptance checkbox at signup and tested it. The refund policy follows the refund work of Member 3 (19 and 20 August) and the live-mode checklist for paid contests that Member 3 wrote on 16 September, and the webcam consent text matches the consent page that I built on 11 August and the consent screen of Member 5. Member 3 filed the written confirmation about paid contests on 17 September.

6.4 Code Freeze and Launch Week (21 to 25 September)

The code freeze and final checks were on 21 September. Member 3 deployed to production on 22 September, the day I watched signups and OTP delivery. On 23 September I watched login errors, and on 24 September I monitored the first real tournament; Member 3’s log records that the small tournament with friends was run on 23 September and that the payouts were checked on 24 September. Member 4 monitored locks and snapshots during

the event (22 and 23 September). On 25 September the whole team held its retro and I noted what to build next.

6.5 After Launch (28 September to 2 October)

On 28 September I reviewed the signup and login errors from the first tournament and on 29 September I fixed small account and room bugs. On 30 September I re-ran the end-to-end test script. On 1 October I updated `api-contract.md` for accounts and rooms, so that the contract matches the code as launched, and on 2 October I reviewed my teammates' pull requests.

6.6 Final Report and Submission (5 and 6 October)

On 5 October I wrote the notes for the final project report, and on 6 October I did the final review and the submission preparation, which ended the logged work.

Table 6.1: Work of weeks 10 to 14

Week	Dates	Work of Member 1
10	07-09 to 11-09	End-to-end script written, extended and run; keyboard and screen-reader checks
11	14-09 to 18-09	Terms, privacy, refund and webcam consent text; acceptance checkbox at signup
12	21-09 to 25-09	Code freeze; monitoring of signups, OTP, login errors and the first real tournament; retro
13	28-09 to 02-10	Error review, small bug fixes, end-to-end script re-run, `api-contract.md`, pull request reviews
14	05-10 to 06-10	Notes for the final report; final review and submission preparation

7. COLLABORATION WITH OTHER MEMBERS

7.1 Overview of Dependencies and Joint Functions

The plan names the pairs that depend on each other: rooms (Member 1) need the judge (Member 2), tournaments (Member 3) need accounts (Member 1) for payout details, and every page needs the design system (Member 5). The plan also asks module owners to write the request and response of a shared API into a contract file before connecting. Figure 7.1 shows my collaboration map.

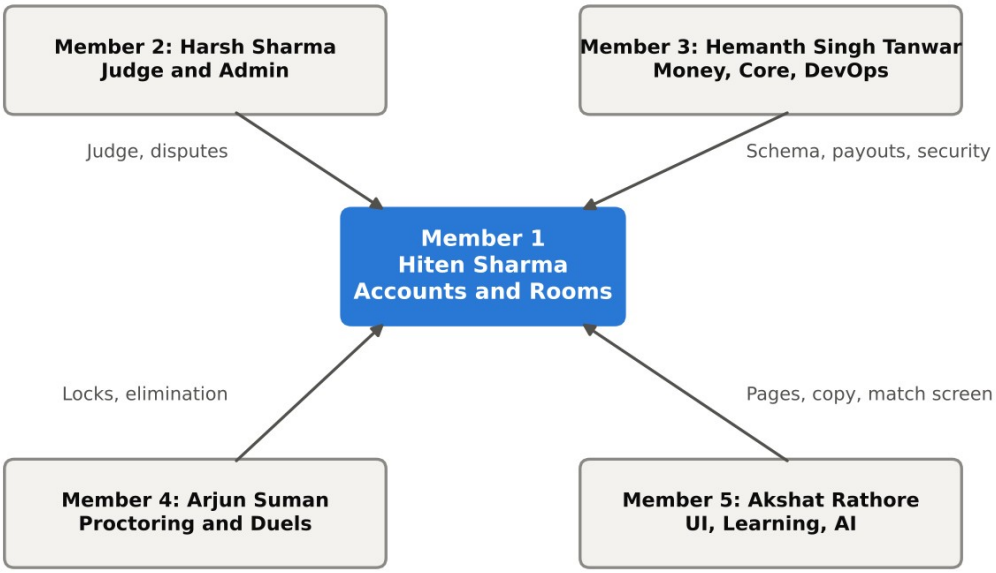


Figure 7.1: Collaboration map of Member 1 with the other four members

Table 7.1: Matrix of joint functions

Joint function	Member 1 side	Other side
Judging a submission	Submission endpoint and on-time check	Member 2: judge, verdicts, concurrency limits
Data model and schema	Room and match model	Member 3: schema V1, migrations, repo layer
Payout details	OTP to the profile mobile	Member 3: payout details and account validation
Final rank list and payouts	Sends the rank list after confirmation	Member 3: payout job, 1% fee, ledger
Tournament lock	Match page refuses locked players	Member 4: lock-state endpoint, snapshots, grace period
Elimination	Server logic with reason	Member 4: eliminate button on host dashboard

Joint function	Member 1 side	Other side
Disputes	Result locking waits on disputes	Member 2: dispute form and admin review
Email delivery	OTP mail through `OtpService.java (mail delivery)`	Member 3: move to a transactional provider (11 September)
Legal pages and paid contests	Refund policy, consent text	Member 3: refund logic, live-mode checklist, written confirmation
Launch	Monitoring of signups, OTP and login errors	Members 3 and 4: deploy, payouts, locks and snapshots
Pages and copy	API behaviour and error cases	Member 5: styling, layouts, empty states, consent and lock screens

7.2 Collaboration with Member 2 (Harsh Sharma)

The submission path is the main link. My endpoint calls Member 2's judge and stores the result. Member 2 first built the submit flow, the test-case run and verdict storage (13 to 15 July), then I built the endpoint (29 and 30 July). Member 2 added a limit on concurrent runs (27 July), one pending submission per player per question (28 July) and stored the code for later similarity checks (31 July). Disputes are the second link: a dispute blocks payout until an administrator decides (17 to 20 August), which my result locking respects. The organisation approval screen of 1 September uses the admin area of Member 2. After launch, Member 2 tested the app as a stranger (14 September) and reviewed the disputes from the first event (24 September).

7.3 Collaboration with Member 3 (Hemanth Singh Tanwar)

Member 3 is the closest partner. We reviewed the room data model together on 10 July, and Member 3 fixed schema V1 on 13 July. Member 3's rate limits on login and OTP (20 July), input validation (21 July), CORS rules (22 July), error handler and request logs (23 July) and tests for the auth routes (24 July) harden the routes that I wrote. Member 3 wrote the money API contract with Member 5 and me on 7 August. Payments follow the account rules: entry fee checkout (17 August), join after payment (18 August) and the age rule (19 August) fit together. The payout details of Member 3 use the OTP and the mobile number of my module (24 August). The rank list that I sent on 26 August feeds the payout job of 26 and 27 August, and we tested result locking together on 27 August. The security review of 31 August to 3 September, with tests for SQL injection, session fixation and token misuse, applied directly to my sessions. In week 10 Member 3 moved email from Gmail to a transactional provider (11 September), and in weeks 11 and 12 prepared the live-mode checklist, deployed to production and ran the first real tournament.

7.4 Collaboration with Member 4 (Arjun Suman)

Member 4 owns the proctoring endpoints: the contract (10 July), then snapshot, violation log and lock state (13 to 15 July). My match page refuses locked players on every request (12 August) and the lock check was tested on 13 August. Member 4 set the tab and minimize lock on the server on 12 August and added the 30 second grace period on 13 August. The host dashboard of Member 4 (17 to 20 August) shows the live player list and the eliminate button, which calls my elimination logic. In weeks 10 to 12 Member 4 tested refresh, two tabs, an unplugged camera and a slow network (7 to 11 September), ran three mock matches (14 to 16 September) and monitored the locks and snapshots during the first event (22 and 23 September).

7.5 Collaboration with Member 5 (Akshat Rathore)

Member 5 gave the pages their look and wording. The login, signup and verify-otp pages (17 July) show the OTP flows of week 2. Member 5 designed create-room and its filters (20 and 21 July) and the Arena cards (22 July), built the match page with the editor, timer, polling scoreboard and countdown (27 to 30 July), and agreed the hook in `PlayController.java for saving wrong submissions with me on 3 August, because PlayController.java` is shared. The consent, camera check and lock screens (10 to 12 August) match my consent page and lock refusal. The money screens of 24 to 28 August use the results that I lock. In weeks 10 to 14 Member 5 worked on contrast and focus states (7 to 9 September), the final index page (14 September) and the feedback of the first players (24 and 28 September).

7.6 Team-Wide Activities

All members follow the same working rhythm from the plan: a short Monday planning meeting, a Wednesday stand-up, a Friday demo and a retro. Main is always deployable, changes merge through a pull request that one other member reviews, and shared files such as `schema.sql`, `HackAgonApplication.java` and `api.js` are changed through Member 3 and Member 5. The whole team held a retro on 25 September and reviewed each other’s pull requests on 2 October, and every member’s log ends with the final review and submission preparation on 6 October.

7.7 Summary of Collaboration

My modules sit at both ends of a contest. Accounts open the door and rooms run the match; everything in between is supplied by others: the judge by Member 2, the data and money by Member 3, the locks by Member 4 and the screens by Member 5. The dates in Appendix B show that each side delivered its part before the other side needed it. Figure 7.2 counts the dated joint work items of Appendix B for each partner, split between the build weeks and

the hardening and launch weeks. Member 3 is the busiest partner, which matches the shared data model, security, payouts and deployment.

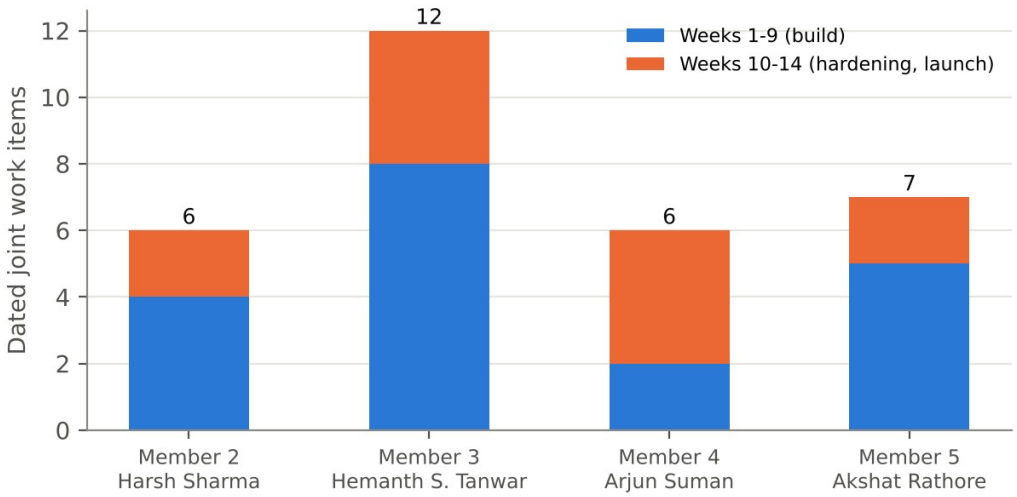


Figure 7.2: Joint work items per partner (from Appendix B)

8. TESTING AND VERIFICATION

8.1 Test Strategy

The plan's definition of done says a feature must work in the browser, have a test or smoke check, show no fake numbers, have hand-written empty and error messages, and be merged to main. I followed this by pairing nearly every build week with a dedicated testing day and a bug-fix day.

8.2 Tests Recorded in the Work Log

Table 8.1: Tests carried out by Member 1

Date	Test	What it checked
24-07-2026	Room creation test	Validation, defaults and form connection; bugs fixed the same day
31-07-2026	Full solo room with two users	The Milestone 1 scenario on the join, submit, judge and scoreboard path
06-08-2026	New account shows zero everywhere	Profile counters come from real records
13-08-2026	Lock check on every request	Locked players are refused by the match page
20-08-2026	Elimination and age rule	No rejoin, ranking removal, 18+ for paid tournaments
27-08-2026	Result locking with Member 3	Confirm, auto-confirm and delivery of the rank list to payouts
04-09-2026	Organisation flow end to end	Application, approval, higher limits and profile
07-09 to 09-09-2026	End-to-end script	Signup, room creation, join, solving a question and results; fixes on 09-09
10-09-2026	Keyboard check	Auth pages can be used with the keyboard
11-09-2026	Screen-reader check	Room pages work with a screen reader
18-09-2026	Acceptance checkbox	Signup with the terms and privacy checkbox
30-09-2026	End-to-end script re-run	The whole path still works after the first tournament

8.3 Bug-fix Days

Bug-fix entries appear on 7 August (bugs found in testing), 14 August (issues found in testing), 21 August (bugs fixed and code reviewed), 28 August (bugs from testing), 9 September (fixes after the first run of the end-to-end script) and 29 September (small account and room bugs after the first tournament). Room creation bugs were also fixed on the testing day of 24 July. The log does not list each bug, so this report does not either.

8.4 Verification by Other Members

Member 3’s tests for the auth routes (24 July), the test checklist before Milestone 1 (31 July) and the security tests (31 August to 3 September) covered code written by me. Member 2’s stranger test (14 September) went from signup to match, and Member 4’s mock matches (14 to 16 September) and the first real tournament (23 September) exercised my room, join and lock code under real use.

8.5 Points to Note

- The tie-break order and the full set of PRD defaults are in the code, but the log does not spell them out. Author to add both in Sections 5.3 and 5.8.
- The log records that the keyboard and screen-reader checks were done but not what they found.
- Email now goes through a transactional provider; OTP delivery was watched on launch day (22 September) and no change to the OTP rules was needed.

9. WEEK-BY-WEEK SUMMARY

Table 9.1 gives the week-by-week summary. Weeks 1 to 13 have 14.5 hours each and week 14 has 5.5 hours.

Table 9.1: Weekly summary of Member 1's work

Week	Dates (log)	Work	Hours
1	06-07 to 10-07	Audit of auth, cookie flags, gaps list; room and match data model reviewed with Member 3	14.5
2	13-07 to 17-07	Email OTP, attempts and resend, SMS OTP, forgot password, profile re-verification	14.5
3	20-07 to 24-07	Room create API, PRD defaults, create-room page connected, status timer, testing	14.5
4	27-07 to 31-07	Join by Arena or code, start match, submission endpoint, judge call, on-time check, solo room test	14.5
5	03-08 to 07-08	Final ranks and tie-break, results data, profile counters, zero-account test, bug fixes	14.5
6	10-08 to 14-08	Tournament mode option, consent page, locked players refused, lock test, bug fixes	14.5
7	17-08 to 21-08	Host elimination, no rejoin, 18+ age rule, tests, bug fixes and review	14.5
8	24-08 to 28-08	Confirm results, 24 hour auto-confirm, rank list to money module, joint test, bug fixes	14.5
9	31-08 to 04-09	Organisation application, approval, limits, final profile page, end-to-end test	14.5
10	07-09 to 11-09	End-to-end script written, extended and run; keyboard and screen-reader checks	14.5
11	14-09 to 18-09	Terms of service, privacy, refund policy, webcam consent text, acceptance checkbox	14.5
12	21-09 to 25-09	Code freeze, monitoring of signups, OTP, login errors and the first real tournament, retro	14.5
13	28-09 to 02-10	Error review, small bug fixes, script re-run, api-contract.md, pull request reviews	14.5
14	05-10 to 06-10	Notes for the final report, final review and submission preparation	5.5
		Total	**194.0**

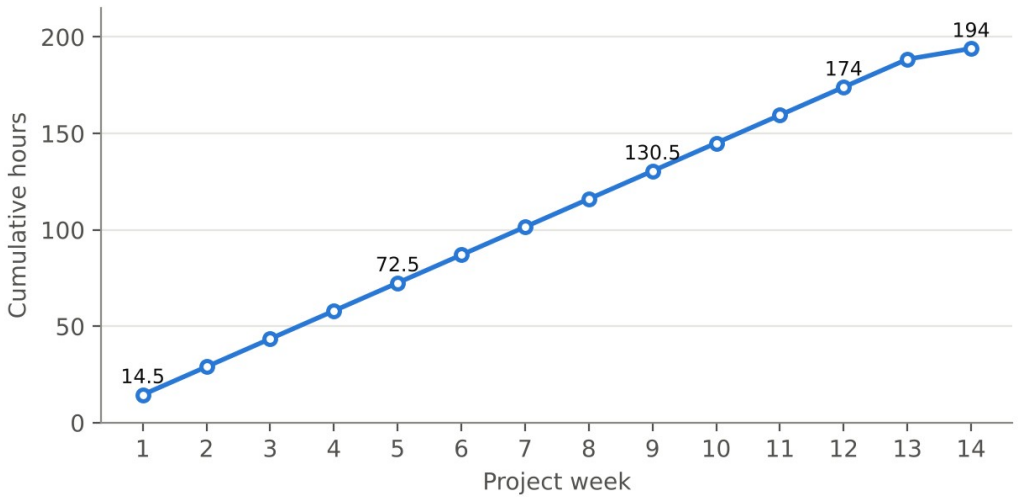


Figure 9.1: Cumulative hours logged by Member 1 week by week

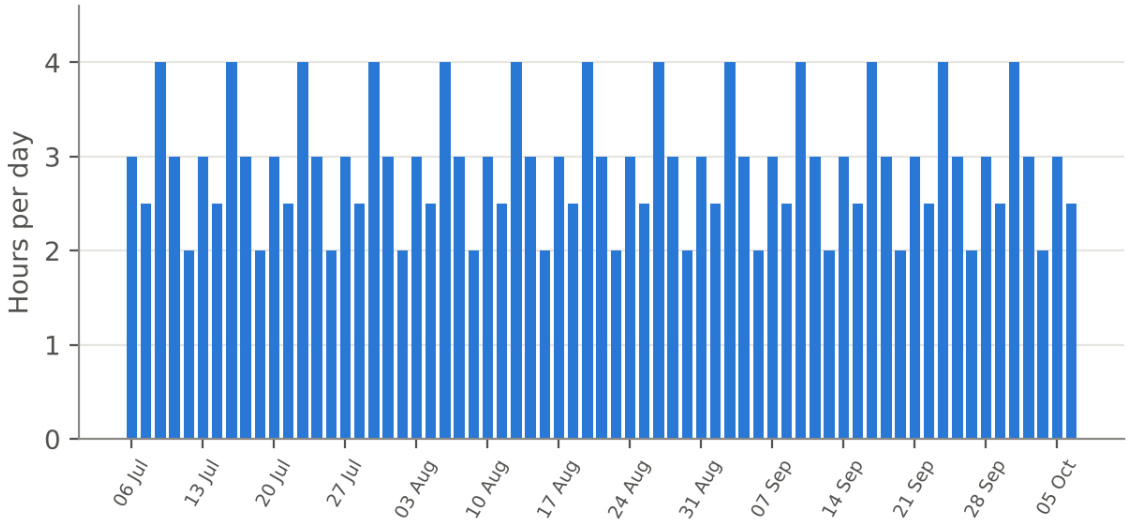


Figure 9.2: Hours logged per day, 6 July to 6 October 2026 (one bar per weekday)

9.1 Weeks 1 and 2: Audit and Accounts

The first week produced the audit and the data model. The second week produced the whole OTP family: email OTP, attempt limits, resend timing, SMS through MSG91, forgot password and profile re-verification.

9.2 Weeks 3 to 5: Rooms, the First Playable Match and Results

Week 3 delivered room creation. Week 4 delivered join, start, submission and judging, ending with the solo room test on 31 July, which is the first milestone of the plan. Week 5 gave the platform ranks, results and honest counters.

9.3 Weeks 6 to 9: Tournament Mode, Host Controls, Result Locking and Organisations

Week 6 added the tournament option and the rule that locked players are refused. Week 7 gave hosts the right to eliminate a player and enforced the age rule. Week 8 closed the loop between a finished match and a payout: confirmation, auto-confirmation and the rank list. Week 9 added organisation hosting and finished the profile page, with a full test of the flow.

9.4 Weeks 10 to 12: Hardening, Legal Pages and Launch

Week 10 gave the project its end-to-end script and the accessibility checks. Week 11 produced the legal pages and the signup checkbox. Week 12 began with the code freeze, followed by launch monitoring and the first real tournament.

9.5 Weeks 13 and 14: After Launch and Handover

Week 13 reviewed the errors of the first tournament, fixed small bugs, re-ran the end-to-end script and brought `api-contract.md` up to date. Week 14 prepared the notes for this report and the final submission.

10. CHALLENGES, DECISIONS AND LIMITATIONS

10.1 Challenges and How They Were Met

Table 10.1: Challenges and responses

Challenge	Response
Existing code did not match the PRD	Audited auth first and listed the gaps before building new features
OTP can be abused by guessing or flooding	5 attempt limit, 5 minute expiry and 60 second resend
Players or hosts could try to change time or status from the browser	Status timer and on-time check run on the server
A paid tournament needs trustworthy age and contact details	18+ rule and re-verification of email and mobile changes
Locks set by one module must be respected by another	Match page refuses locked players on every request, tested on 13 August
Results must not be changed after payouts begin	Host confirmation or 24 hour auto-confirm before the rank list is sent
Shared files can cause merge conflicts	Changes to `PlayController.java` agreed with Member 5; schema changes go through Member 3
Users must accept legal terms before a paid contest	Terms, privacy, refund and webcam consent text, with an acceptance checkbox at signup (14 to 18 September)
The first real tournament could expose errors that tests miss	Monitored signups, OTP delivery and login errors; reviewed and fixed what appeared (22 to 29 September)

10.2 Delivered Features

Table 10.2: Status of Member 1’s features

Feature	Status
Email OTP, SMS OTP, forgot password, profile re-verification	Delivered (log, 13 to 17 July)
Room create, defaults, status timer	Delivered (log, 20 to 24 July)
Join, start, submission endpoint, on-time check	Delivered (log, 27 to 31 July)
Results, tie-break, profile counters	Delivered (log, 3 to 7 August)
Tournament mode, consent, lock refusal	Delivered (log, 10 to 14 August)
Elimination, age rule	Delivered (log, 17 to 21 August)
Result locking and rank list	Delivered (log, 24 to 28 August)
Organisation hosting	Delivered (log, 31 August to 4 September)
End-to-end script, keyboard and screen-reader check	Delivered (log, 7 to 11 September)
Legal pages and signup acceptance checkbox	Delivered (log, 14 to 18 September)

Feature	Status
Launch monitoring and first tournament	Delivered (log, 21 to 25 September)
Post-launch fixes, API contract and report preparation	Delivered (log, 28 September to 6 October)

10.3 Notes on the Sources

- The log describes each task in one line. Details such as the tie-break order, the exact PRD default values and the OTP table design are not stated in either document, so they are not invented here.
- Dates and hours are those of the log. They are the only source for what was done on each day.
- The plan lists twelve weeks of tasks; the log covers fourteen project weeks, and the last two are the review after launch and the preparation of this report.

10.4 Limitations

The report relies on the plan and the log and does not contain test output or code listings. Hours are as recorded in the log. Before submission I should fill in the two highlighted points (the tie-break order and the PRD default values).

11. CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

Over fourteen logged weeks and 194.0 hours, I delivered the accounts and OTP module and the rooms and matches module of HackAgon and took them through testing, legal pages and launch. A user can sign up and verify by email or SMS, recover a password, edit a profile with re-verification, accept the terms, create a room with sensible editable defaults, join by the Arena or an invite code, submit code that is judged and scored on time, see ranks and honest counters, and play in tournament mode under locks. Hosts can eliminate players and confirm results, and organisations can apply for higher limits. The first real tournament ran on 23 September, and the weeks after it were used to review the errors and fix small bugs. Each task is traced to a dated log entry and to the partner module it connects with.

11.2 Lessons Learnt

- Auditing before building saves time, because the existing skeleton already covered part of the work.
- Server-side rules for time, locks and results are the base of fairness in any contest platform.
- A shared OTP library keeps signup, reset and profile change consistent, and let the mail provider change without touching the rules.
- Early agreement on interfaces, such as the rank list for the money module, lets two members finish within days of each other.
- A testing day and a bug-fix day after each build week keep the main branch deployable.
- A scripted end-to-end test is cheap to write and was worth re-running after the first real event.

11.3 Future Scope

- Add rate-based monitoring of OTP delivery and login errors, since both were checked by hand in launch week.
- Run the end-to-end script automatically on every deploy.
- Record the findings of the keyboard and screen-reader checks and repeat them for new pages.
- Support more room formats and finer host controls once duos, squads and duels are used heavily.
- Have the terms of service, privacy policy and refund policy reviewed by a legal adviser before wider use.

REFERENCES

[1] HackAgon Team Modules and 12-Week Plan, HackAgon team, 2026.

[2] HackAgon Daily Work Log, 6 July 2026 to 6 October 2026, HackAgon team.

[3] Java SE 17 documentation, docs.oracle.com.

[4] Spring Boot reference documentation, docs.spring.io.

[5] MySQL Reference Manual, dev.mysql.com.

[6] MSG91 documentation, msg91.com.

[7] JJWT documentation, github.com/jwtk/jjwt.

APPENDIX A: COMPLETE DAILY WORK LOG OF MEMBER 1

The log below is reproduced from the HackAgon Daily Work Log, Member 1 (Accounts, Rooms), 6 July 2026 to 6 October 2026. Total: 67 days, 194.0 hours.

Table A.1: Daily work log of Member 1

Date	Hrs	Status	Work description
Week 1			
06-07-2026	3	Completed	Read AuthController.java and SecurityConfig.java, JwtService.java to understand login flow
07-07-2026	2.5	Completed	Checked password hashing and JWT expiry settings
08-07-2026	4	Completed	Checked cookie flags and listed gaps against the PRD
09-07-2026	3	Completed	Drafted room and match data model
10-07-2026	2	Completed	Reviewed data model with Member 3 and noted changes
Week 2			
13-07-2026	3	Completed	Built email OTP with 5 minute expiry
14-07-2026	2.5	Completed	Added 5 attempt limit and 60 second resend
15-07-2026	4	Completed	Added SMS OTP using MSG91
16-07-2026	3	Completed	Built forgot password flow with OTP
17-07-2026	2	Completed	Built profile edit that asks for new OTP on email or mobile change
Week 3			
20-07-2026	3	Completed	Built room create API with validation
21-07-2026	2.5	Completed	Added editable PRD default values to room form
22-07-2026	4	Completed	Connected create-room.html to the API
23-07-2026	3	Completed	Added server timer for room status open, live, finished
24-07-2026	2	Completed	Tested room creation and fixed bugs
Week 4			
27-07-2026	3	Completed	Built join room by Arena and invite code
28-07-2026	2.5	Completed	Added start match logic
29-07-2026	4	Completed	Built submission endpoint in PlayController.java
30-07-2026	3	Completed	Added judge call and server-side on-time check
31-07-2026	2	Completed	Tested a full solo room with two users
Week 5			
03-08-2026	3	Completed	Added final ranks and tie-break rule
04-08-2026	2.5	Completed	Built results page data

Date	Hrs	Status	Work description
05-08-2026	4	Completed	Calculated profile counters from real records
06-08-2026	3	Completed	Tested that a new account shows zero everywhere
07-08-2026	2	Completed	Fixed bugs found in testing
Week 6			
10-08-2026	3	Completed	Added tournament mode option to room creation
11-08-2026	2.5	Completed	Built consent page before joining
12-08-2026	4	Completed	Made match page refuse locked players
13-08-2026	3	Completed	Tested lock check on every request
14-08-2026	2	Completed	Fixed issues found in testing
Week 7			
17-08-2026	3	Completed	Built host eliminate player with reason
18-08-2026	2.5	Completed	Blocked rejoin and removed eliminated score from ranking
19-08-2026	4	Completed	Added 18+ age rule for paid tournaments
20-08-2026	3	Completed	Tested elimination and age rule
21-08-2026	2	Completed	Fixed bugs and reviewed code
Week 8			
24-08-2026	3	Completed	Built host confirm results button
25-08-2026	2.5	Completed	Added 24 hour auto-confirm with no dispute
26-08-2026	4	Completed	Sent final rank list to the money module
27-08-2026	3	Completed	Tested result locking with Member 3
28-08-2026	2	Completed	Fixed bugs from testing
Week 9			
31-08-2026	3	Completed	Built organization host application form
01-09-2026	2.5	Completed	Built admin approval flow for organizations
02-09-2026	4	Completed	Added higher limits for approved organizations
03-09-2026	3	Completed	Finalised profile page
04-09-2026	2	Completed	Tested organization flow end to end
Week 10			
07-09-2026	3	Completed	Wrote script for signup, room creation and join
08-09-2026	2.5	Completed	Extended script to solve a question and show results
09-09-2026	4	Completed	Ran the script and fixed what broke
10-09-2026	3	Completed	Did keyboard check of auth pages
11-09-2026	2	Completed	Did screen-reader check of room pages
Week 11			

Date	Hrs	Status	Work description
14-09-2026	3	Completed	Wrote terms of service
15-09-2026	2.5	Completed	Wrote privacy policy
16-09-2026	4	Completed	Wrote refund policy
17-09-2026	3	Completed	Wrote webcam consent text
18-09-2026	2	Completed	Added acceptance checkbox at signup and tested
Week 12			
21-09-2026	3	Completed	Code freeze and final checks
22-09-2026	2.5	Completed	Watched signups and OTP delivery
23-09-2026	4	Completed	Watched login errors
24-09-2026	3	Completed	Monitored the first real tournament
25-09-2026	2	Completed	Held team retro and noted what to build next
Week 13			
28-09-2026	3	Completed	Reviewed signup and login errors from first tournament
29-09-2026	2.5	Completed	Fixed small account and room bugs
30-09-2026	4	Completed	Re-ran the end-to-end test script
01-10-2026	3	Completed	Updated api-contract.md for accounts and rooms
02-10-2026	2	Completed	Reviewed teammates' pull requests
Week 14			
05-10-2026	3	Completed	Wrote notes for the final project report
06-10-2026	2.5	Completed	Final review and submission preparation

APPENDIX B: DATED JOINT WORK WITH OTHER MEMBERS

Each row pairs a task of Member 1 with the task of another member that it depends on or supports, as recorded in the daily log.

Table B.1: Dated joint work

Member 1 task	Other member task	With
09-07 and 10-07: room and match model drafted and reviewed	07-07 schema reviewed against PRD; 13-07 schema V1 tables fixed	Member 3
13-07 to 17-07: OTP flows	20-07 rate limits on login and OTP; 24-07 tests for auth routes	Member 3
13-07 to 17-07: OTP flows	17-07 login, signup and verify-otp pages styled	Member 5
22-07: create-room.html connected	24-07 filter API (Member 2); 20-07 and 21-07 create-room page and filters (Member 5)	Member 2, Member 5
29-07 and 30-07: submission endpoint and judge call	13-07 to 15-07 submit, test cases, verdicts; 27-07 and 28-07 concurrency and pending limits; 31-07 code stored	Member 2
31-07: full solo room with two users	30-07 Milestone 1 deployed; 31-07 test checklist	Member 3
04-08 to 06-08: results and counters	27-07 to 31-07 match.html, scoreboard, countdown; 03-08 hook in PlayController.java agreed	Member 5
07-08: money API contract	07-08 contract written with Members 1 and 5	Member 3
11-08: consent page	10-08 to 12-08 consent, camera check and lock screens designed	Member 5
12-08 and 13-08: lock refusal and test	15-07 lock-state endpoint; 12-08 tab and minimize lock; 13-08 grace period	Member 4
17-08 and 18-08: elimination logic	17-08 live player list; 20-08 eliminate button	Member 4
19-08: age rule for paid tournaments	17-08 entry fee checkout; 18-08 join after payment	Member 3
24-08 and 25-08: confirm and auto-confirm	17-08 to 20-08 dispute form, payout block, admin review	Member 2
26-08 and 27-08: rank list sent and joint test	26-08 payout job with 1% fee; 27-08 split by rank and leftover paise	Member 3
17-07 and 24-08: profile mobile change needs OTP	24-08 payout details with OTP to profile mobile	Member 3
01-09: organization approval flow	03-08 to 07-08 admin login protection and admin screens	Member 2
01-09 to 03-09: security review of sessions	01-09 SQL injection test; 02-09 session fixation and token misuse; 03-09 fixes	Member 3
04-09: organization flow end to end	24-08 to 28-08 money screens, winner screen	Member 5

Member 1 task	Other member task	With
07-09 to 09-09: end-to-end script	14-09 Member 2 tested the app as a stranger, signup to match	Member 2
10-09 and 11-09: keyboard and screen-reader checks	07-09 to 09-09 colour contrast and focus states	Member 5
10-09 to 11-09: auth emails	11-09 email moved from Gmail to a transactional provider	Member 3
16-09: refund policy	17-09 written confirmation about paid contests; 16-09 Razorpay live-mode checklist	Member 3
17-09: webcam consent text	07-09 to 17-09 proctoring tests and mock matches; 18-09 host guide for tournament mode	Member 4
22-09 to 24-09: monitoring signups, OTP, login errors and the first tournament	22-09 deploy to production; 23-09 first real tournament; 24-09 payouts checked	Member 3
22-09 to 24-09: monitoring	22-09 and 23-09 locks and snapshots monitored during the event	Member 4
24-09: first tournament	24-09 disputes from the first event reviewed	Member 2
28-09 to 29-09: account and room bugs	29-09 lock rules tuned; 29-09 small UI issues fixed from feedback	Member 4, Member 5
01-10: api-contract.md updated for accounts and rooms	07-08 money API contract (Member 3); 10-07 proctor API contract (Member 4)	Member 3, Member 4